

Monday 3 – Classification Investigation

Individual Investigation

1. Question

Why did the Decision Tree give better accuracy than Logistic Regression on the same dataset?

2. Investigation

I used the same target, features, train/test split, and median imputation for both models. I compared their test accuracy and confusion matrices to understand where the difference came from.

3. Evidence

Logistic Regression

Accuracy: 73.3%

Confusion matrix:

```
[[4, 3],  
 [5, 18]]
```

Decision Tree

Accuracy: 80.0%

Confusion matrix:

```
[[4, 3],  
 [3, 20]]
```

The Decision Tree made two fewer mistakes than Logistic Regression. Both models correctly predicted 4 cases from the first class. For completed quizzes, the Decision Tree correctly predicted 20 cases, while Logistic Regression correctly predicted 18.

So, even though the accuracy difference looks noticeable (73.3% vs. 80.0%), the actual difference was only two predictions in a test set of 30 rows.

4. Conclusion

The Decision Tree may handle threshold-based or non-linear patterns in the data better than Logistic Regression.

However, the test set contained only 30 rows, and the difference between the models was only two predictions. Because of the small test set, I cannot conclude that the Decision Tree is clearly better.

Testing with different train/test splits or cross-validation would help check whether this difference is consistent or mainly caused by this particular split.

Reference

Monday 3 – Classification Investigation Task:

[Github link for reference](#),